



Part 1: Understanding Security Assessment and VAPT Fundamentals

1. Introduction

- Security Assessment and VAPT (Vulnerability Assessment and Penetration Testing) are important processes used to identify weaknesses in systems before attackers can exploit them. Instead of waiting for a security incident to happen, organizations perform these assessments to understand their security posture and reduce risk. In this task, the focus is on understanding how security assessments are carried out, how VAPT is performed in a structured way, and how risks and findings are properly documented.

1.1. Understanding Security Assessment

- A security assessment is the process of reviewing systems, networks, and applications to find potential security gaps. The main objective is to identify vulnerabilities and understand how they can impact the system if left unresolved.
- Security assessments are generally categorized into different types:
 - **Security Assessment:**
This focuses on identifying security weaknesses by following standard frameworks such as NIST. It helps organizations understand where their security controls are lacking.
 - **Vulnerability Assessment:**
In this approach, automated tools like OpenVAS are used to scan systems for known vulnerabilities, misconfigurations, and outdated services without actively exploiting them.
 - **Penetration Testing:**
Penetration testing simulates real-world attacks using tools such as Nmap and Metasploit. It helps verify whether the identified vulnerabilities can actually be exploited by an attacker.
 - **Compliance Testing:**
This ensures that systems follow security standards and best practices, such as CIS Benchmarks, to meet regulatory and organizational requirements.

1.2. VAPT Methodology

- VAPT is performed using a structured methodology to ensure testing is systematic and results are reliable. The methodology consists of the following phases:
 - **Planning:**
In this phase, the scope of testing is defined, including target systems, IP addresses, and tools to be used.
 - **Discovery:**
Information is collected using tools like Nmap for network scanning and other scanners to identify open ports, services, and potential entry points.



- **Attack:**
Identified vulnerabilities are validated through controlled exploitation using penetration testing tools.
 - **Reporting:**
All findings are documented clearly, including vulnerability details, screenshots, severity levels, and remediation suggestions.
- This approach ensures that security testing is organized and easy to understand.

1.3. Security Standards and Compliance

- Security standards provide guidelines for protecting systems and sensitive data. Standards such as ISO 27001, GDPR, and HIPAA help organizations maintain strong security controls. During VAPT activities, frameworks like the OWASP Top 10 are commonly used to prioritize critical vulnerabilities and focus on high-risk issues.

1.4. Risk Assessment Basics

- Risk assessment is used to prioritize vulnerabilities based on their severity. The CVSS (Common Vulnerability Scoring System) is commonly used to calculate vulnerability scores. Based on these scores, risks are categorized as High, Medium, or Low, which helps in deciding which issues need immediate attention.

1.5. Common Vulnerabilities

- Some common vulnerabilities identified during assessments include open ports, weak authentication, outdated services, and web application issues such as SQL Injection and Cross-Site Scripting (XSS). Practicing on vulnerable environments like Metasploitable helps in understanding how these vulnerabilities exist and how they can be exploited.

1.6. Importance of Documentation

- Documentation is an essential part of VAPT. Proper documentation ensures that findings are clearly communicated and can be easily understood by stakeholders. A good report includes vulnerability details, evidence screenshots, risk ratings, and recommended fixes.

Part 2: Testing Environment and Tools Used

2. Testing Environment Setup

- For this VAPT task, a controlled virtual lab environment was created to safely perform vulnerability assessment and penetration testing activities. The environment was designed to simulate a real-world testing scenario while ensuring that no live or production systems were affected during testing.
- The testing setup included one attacker machine and one vulnerable target machine, both connected on the same virtual network to allow proper communication.



Figure 1: Virtual machines running (Kali Linux and Metasploitable 2)

2.1 Attacker Machine – Kali Linux

- Kali Linux was used as the attacker machine for performing all VAPT activities. Kali Linux is a widely used penetration testing operating system that comes with pre-installed security assessment tools.
- The following tasks were performed from the Kali Linux machine:
 - Network scanning
 - Vulnerability identification
 - Penetration testing
 - Evidence and screenshot collection
- Kali Linux provided a stable and suitable platform for carrying out all security testing operations.

```
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:ce:b5:57 brd ff:ff:ff:ff:ff:ff
    inet 192.168.177.130/24 brd 192.168.177.255 scope global dynamic noprefixroute eth0
        valid_lft 1092sec preferred_lft 1092sec
```

Figure 2: Kali Linux terminal showing IP address configuration

2.2 Target Machine – Metasploitable 2

- Metasploitable 2 was used as the target system for testing. It is an intentionally vulnerable virtual machine designed specifically for practicing vulnerability assessment and penetration testing techniques.
- The target system contains multiple known vulnerabilities, making it suitable for learning and demonstration purposes. Before starting the assessment, the target machine's IP address was identified and verified.



```
No mail.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:ac:03:18
          inet addr:192.168.177.134  Bcast:192.168.177.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:feac:318/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:39 errors:0 dropped:0 overruns:0 frame:0
          TX packets:68 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4249 (4.1 KB)  TX bytes:7220 (7.0 KB)
          Interrupt:19 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:96 errors:0 dropped:0 overruns:0 frame:0
          TX packets:96 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:21437 (20.9 KB)  TX bytes:21437 (20.9 KB)

msfadmin@metasploitable:~$ _
```

Figure 3: Metasploitable 2 terminal showing assigned IP address

2.3 Network Configuration

- Both Kali Linux and Metasploitable 2 were configured on the same virtual network using a NAT or Host-Only adapter. This ensured:
 - Secure and isolated testing
 - Proper communication between attacker and target
 - No exposure to external networks
- Network connectivity was verified before proceeding with further testing.

Virtual Machine Settings		Virtual Machine Settings	
Hardware		Hardware	
Device	Summary	Device	Summary
Memory	5 GB	Memory	512 MB
Processors	2	Processors	1
Hard Disk (SCSI)	80.1 GB	Hard Disk (SCSI)	8 GB
Network Adapter	NAT	CD/DVD (IDE)	Auto detect
USB Controller	Present	Network Adapter	NAT
Sound Card	Auto detect	Network Adapter 2	Host-only
Display	Auto detect	USB Controller	Present
		Display	Auto detect

Figure 4: Network adapter settings of both virtual machines

2.4 Connectivity Verification

- To confirm that both machines could communicate properly, connectivity tests were performed using the ping command.
- From Kali Linux, the target system was pinged successfully, confirming that the network setup was correct and ready for assessment.



```
(kali@kali)-[~]
$ ping -c 4 192.168.177.134
PING 192.168.177.134 (192.168.177.134) 56(84) bytes of data.
64 bytes from 192.168.177.134: icmp_seq=1 ttl=64 time=1.21 ms
64 bytes from 192.168.177.134: icmp_seq=2 ttl=64 time=2.91 ms
64 bytes from 192.168.177.134: icmp_seq=3 ttl=64 time=1.12 ms
64 bytes from 192.168.177.134: icmp_seq=4 ttl=64 time=0.653 ms

— 192.168.177.134 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 0.653/1.474/2.912/0.856 ms
```

Figure 5: Successful ping from Kali Linux to Metasploitable 2

2.5 Tools Used

- The following tools were used during this VAPT task:
 - **Nmap** – Used for network discovery and port scanning
 - **OpenVAS** – Used for vulnerability scanning and risk identification
 - **Metasploit Framework** – Used for validating vulnerabilities through controlled exploitation
 - **Wireshark** – Used for network traffic capture and analysis
 - **Basic Utilities** – ping and other system commands for verification
- Tool availability was verified before starting the vulnerability assessment.

```
(kali@kali)-[~]
$ nmap --version
openvas --version
msfconsole --version

Nmap version 7.95 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.7 openssl-3.4.1 libssh2-1.11.1 libz-1.3.1 libpcr2-10.45 libpcap-1.10.5 nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select
OpenVAS 23.13.1
gvm-libs 22.34.2
Most new code since 2005: (C) 2024 Greenbone AG
Nessus origin: (C) 2004 Renaud Deraison <deraison@nessus.org>
License GPLv2: GNU GPL version 2
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Framework Version: 6.4.50-dev
```

Figure 6: Kali Linux showing installed security tools

Part-2 Summary

In this part, the testing environment was successfully set up and verified. Both the attacker and target machines were properly configured, network connectivity was confirmed, and the required tools were ready for use. This environment served as the foundation for the vulnerability scanning and penetration testing activities performed in the next parts of the task.



Part 3: Vulnerability Assessment Using OpenVAS

3. Introduction

- In this part, a vulnerability assessment was performed on the target system using OpenVAS (Greenbone Vulnerability Manager). The objective of this assessment was to identify security vulnerabilities, misconfigurations, and outdated services running on the target machine. The scan results were later used for risk assessment and prioritization.

3.1 Tool Used – OpenVAS

- OpenVAS is an open-source vulnerability scanning tool used to detect known security issues in systems and services. It provides detailed vulnerability information, including CVSS scores, severity levels, affected ports, and recommended remediation steps.
- OpenVAS was selected for this task because it is widely used in vulnerability assessment processes and is suitable for both learning and practical security testing.

3.2 Target Information

- **Target System:** Metasploitable 2
- **Target IP Address:** 192.168.177.134
- **Operating System:** kali linux
- **Scan Type:** Full and Fast
- **Scanner Used:** OpenVAS Default Scanner

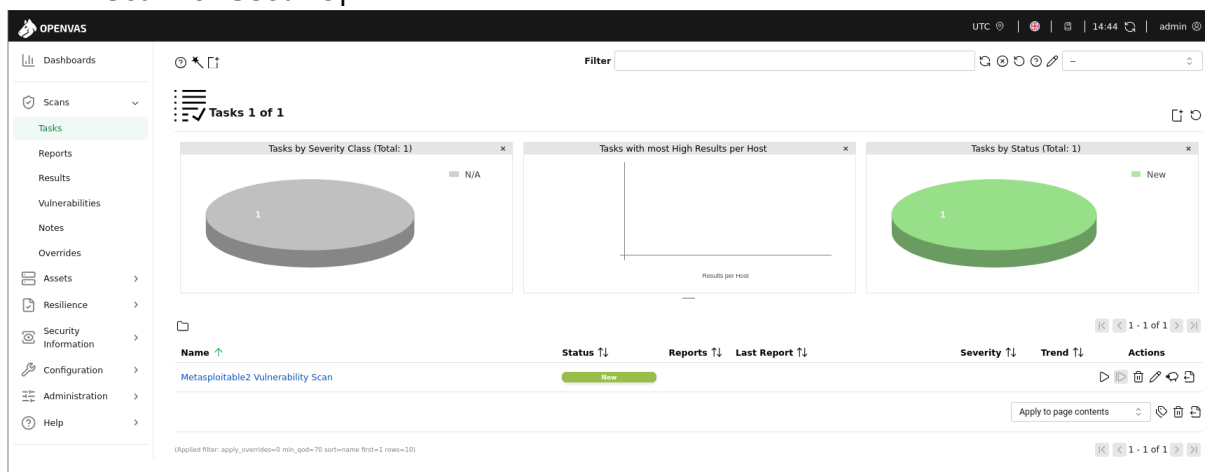


Figure 7: OpenVAS dashboard showing scan task created for Metasploitable 2

3.3 Scan Configuration

- The scan task was configured using the “**Full and Fast**” scan profile to ensure a balance between scan depth and time. The port list was set to “**All TCP and Nmap top 100 UDP**”, allowing OpenVAS to scan all TCP ports and the most commonly used UDP ports.
- This configuration helps identify the majority of critical and high-risk vulnerabilities.

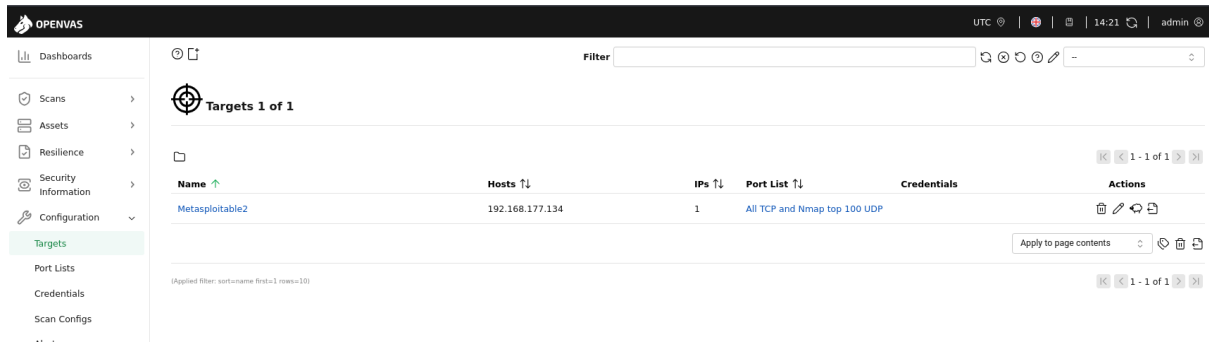


Figure 8: Target configuration showing IP address and port list selection

3.4 Scan Execution

- After configuring the scan task, the scan was started from the OpenVAS interface. The scan progressed through different stages such as requested, running, and completed. The total scan duration depended on the number of services detected on the target system.
- Once completed, the scan status changed to **Done**, and the results were made available for analysis.

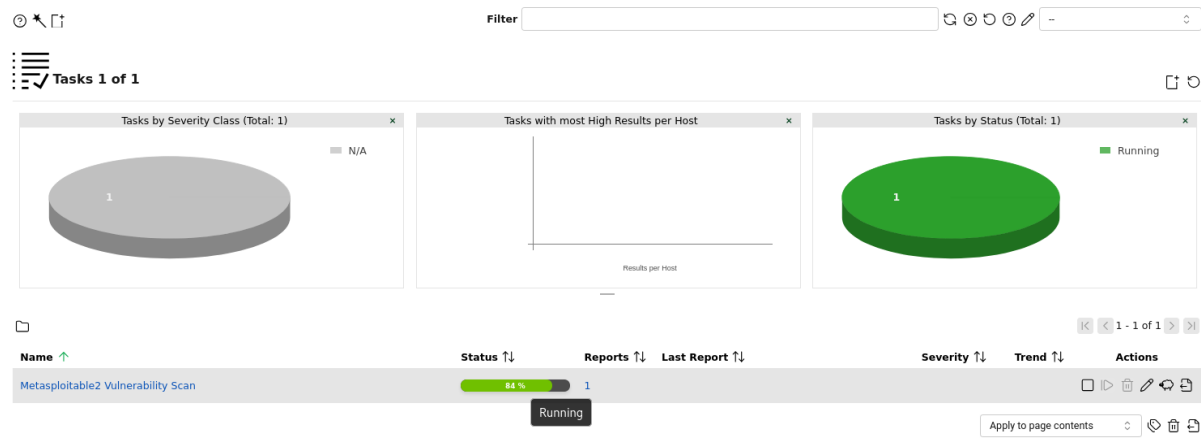


Figure 9: Scan status showing “Running”

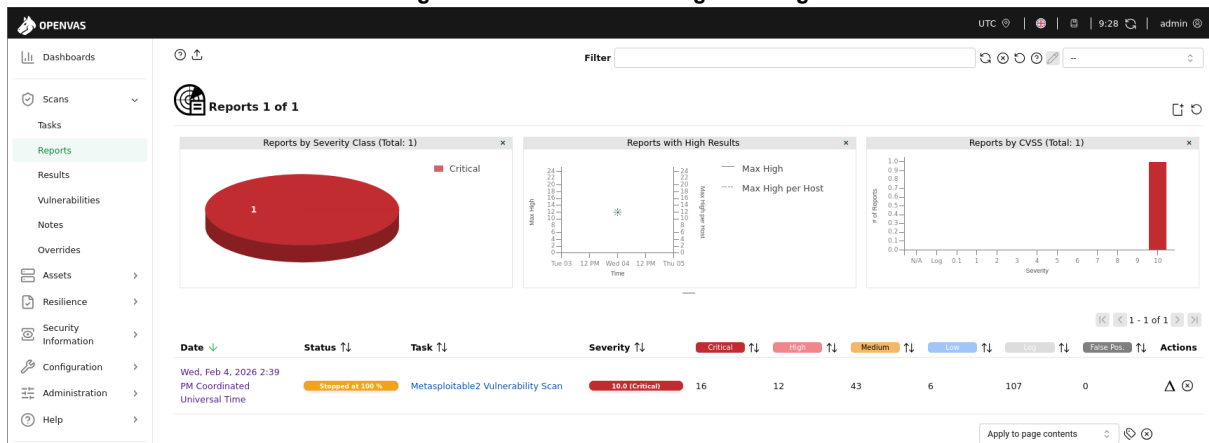


Figure 10: Scan status showing “Completed (Done)”



3.5 Scan Results Overview

- The scan results revealed multiple vulnerabilities across different services running on the target system. Since Metasploitable 2 is intentionally vulnerable, a high number of critical and high-severity findings were expected.
- **Summary of Findings**
 - **Critical Vulnerabilities:** Multiple
 - **High Severity Vulnerabilities:** Detected
 - **Medium Severity Vulnerabilities:** Detected
 - **Low Severity Vulnerabilities:** Detected
- The majority of critical vulnerabilities were related to outdated services and default credentials.

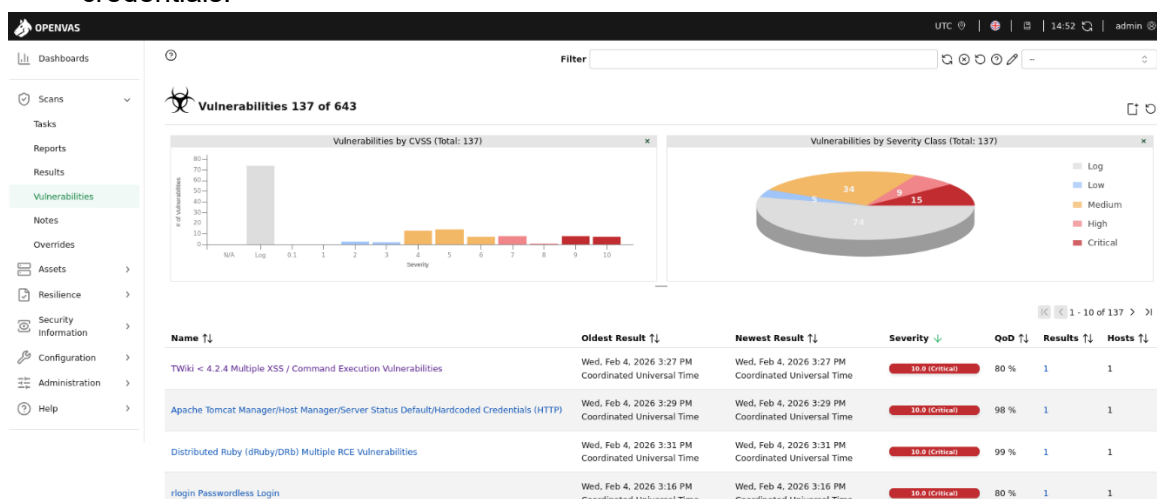


Figure 11: OpenVAS vulnerability summary showing severity distribution

3.6 Key Vulnerabilities Identified

- The following major vulnerabilities were identified during the scan:
- 1. **PHP CGI Remote Code Execution**
 - **Port:** 80 (HTTP)
 - **Severity:** Critical
 - **CVSS Score:** 10.0
 - **Description:** Allows remote attackers to execute arbitrary commands on the server.
 - **Impact:** Full system compromise possible.



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Insight

When PHP is used in a CGI-based setup (such as Apache's mod_cgid), the php-cgi receives a processed query string parameter as command line arguments which allows command-line switches, such as -s, -d or -c to be passed to the php-cgi binary, which can be exploited to disclose source code and obtain arbitrary code execution.

An example of the -s command, allowing an attacker to view the source code of index.php is below:

`http://example.com/index.php?s=`

Detection Method

Send multiple a crafted HTTP POST requests and checks the responses.

Notes:

- This script checks for the presence of CVE-2012-1823 which indicates that the system is also affected by the other included CVEs.
- It is currently expected that a result of this VT is reported if the system is generally exposing a phpinfo() output on the relevant URL / endpoint (independent from the running product). Exposing such sensitive information is generally seen as a security misconfiguration and should be avoided.

Quality of Detection: remote_active (95%)

Affected Software/OS

PHP versions prior to 5.3.13 and 5.4.x prior to 5.4.3.

Other products / applications might be affected by the tested CVE-2012-1823 as well.

Impact

Exploiting this issue allows remote attackers to view the source code of files in the context of the server process. This may allow the attacker to obtain sensitive information and to run arbitrary PHP code on the affected computer. Other attacks are also possible.

Figure 12: PHP CGI RCE vulnerability details page

2. Apache Tomcat Default Credentials

- **Port:** 8180
- **Severity:** Critical
- **Description:** Default login credentials allow unauthorized access to the Tomcat management interface.
- **Impact:** Attacker can deploy malicious applications or gain server access.

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NVT: Apache Tomcat Manager/Host Manager/Server Status Default/Hardcoded Credentials (...)

Information Preferences User Tags

Summary

The Apache Tomcat Manager/Host Manager/Server Status is using default or known hardcoded credentials.

Scoring

CVSS

CVSS Base 10.0 (Critical)

CVSS Base Vector AV:N/AC:L/Au:N/C:C/CIA:C

CVSS Origin N/A

CVSS Date Wed, Aug 22, 2012 3:19 PM Coordinated Universal Time

Detection Method

Quality of Detection: remote_app (98%)

Impact

An attacker can exploit this issue to upload and execute arbitrary code, which will facilitate a complete compromise of the affected computer.

Figure 13: Tomcat default credentials vulnerability details

3. MySQL Default Credentials

- **Port:** 3306
- **Severity:** Critical
- **Description:** Database access possible using default credentials.
- **Impact:** Unauthorized access to sensitive data.



The screenshot shows the OpenVAS web interface. On the left is a sidebar with navigation links: Dashboards, Scans, Assets, Resilience, Security Information, NVTs (selected), CVEs, CPEs, CERT-Bund Advisories, DFN-CERT Advisories, Configuration, Administration, and Help. The main content area displays details for the NVT 'MySQL / MariaDB Default Credentials (MySQL Protocol)'. It includes tabs for Information, Preferences, and User Tags. The 'Information' tab is active, showing a Summary, Scoring, CVSS details, Detection Method, and Affected Software/OS. The CVSS Base score is 9.8 (Critical). The CVSS Base Vector is CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H. The CVSS Origin is NVD. The CVSS Date is Wed, Feb 7, 2024 5:37 PM Coordinated Universal Time. The Detection Method is remote_active (95%). The Affected Software/OS section lists CVE-2001-0645: Symantec/AXENT NetProwler 3.5.x.

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NVT: MySQL / MariaDB Default Credentials (MySQL Protocol)

Information Preferences (0) User Tags (0)

Summary

It was possible to login into the remote MySQL using default credentials.

Scoring

CVSS

CVSS Base **9.8 (Critical)**

CVSS Base Vector **CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H**

CVSS Origin NVD

CVSS Date Wed, Feb 7, 2024 5:37 PM Coordinated Universal Time

Detection Method

Quality of Detection: remote_active (95%)

Affected Software/OS

The following products are know to use such weak credentials:

- CVE-2001-0645: Symantec/AXENT NetProwler 3.5.x

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Figure 14: MySQL default credentials vulnerability details

4. vsftpd Backdoor Vulnerability

- **Port:** 21 (FTP)
- **Severity:** Critical
- **Description:** A known backdoor vulnerability allowing remote shell access.
- **Impact:** Complete system compromise.

The screenshot shows the OpenVAS web interface. On the left is a sidebar with navigation links: Dashboards, Scans, Assets, Resilience, Security Information, NVTs (selected), CVEs, CPEs, CERT-Bund Advisories, DFN-CERT Advisories, Configuration, Administration, and Help. The main content area displays details for the NVT 'vsftpd backdoor vulnerability'. It includes tabs for Information, Preferences, and User Tags. The 'Information' tab is active, showing a Summary, Scoring, CVSS details, Insight, Detection Method, and Affected Software/OS. The CVSS Base score is 9.8 (Critical). The CVSS Base Vector is CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H. The CVSS Origin is NVD. The CVSS Date is Mon, Apr 12, 2021 7:15 PM Coordinated Universal Time. The Insight section states: 'The tainted source package contains a backdoor which opens a shell on port 6200/tcp.' The Detection Method is remote_vul (99%). The Affected Software/OS section lists: 'The vsftpd 2.3.4 source package downloaded between 20110630 and 20110703 is affected.' The Impact section states: 'Attackers can exploit this issue to execute arbitrary commands in the context of the application. Successful attacks will compromise the affected application.'

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vsftpd backdoor vulnerability

Information Preferences (0) User Tags (0)

Summary

vsftpd is prone to a backdoor vulnerability.

Scoring

CVSS

CVSS Base **9.8 (Critical)**

CVSS Base Vector **CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H**

CVSS Origin NVD

CVSS Date Mon, Apr 12, 2021 7:15 PM Coordinated Universal Time

Insight

The tainted source package contains a backdoor which opens a shell on port 6200/tcp.

Detection Method

Quality of Detection: remote_vul (99%)

Affected Software/OS

The vsftpd 2.3.4 source package downloaded between 20110630 and 20110703 is affected.

Impact

Attackers can exploit this issue to execute arbitrary commands in the context of the application. Successful attacks will compromise the affected application.

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Figure 15: vsftpd backdoor vulnerability details



3.7 Observations

- Most vulnerabilities were **remotely exploitable**.
- Several services were **outdated and unsupported**.
- Default credentials significantly increased attack risk.
- The target system is highly insecure and should never be exposed to a production network.

3.8 Conclusion of Vulnerability Scan

- The OpenVAS scan successfully identified multiple critical vulnerabilities on the target system. These findings highlight the importance of regular vulnerability assessments, timely patching, and secure configuration practices. The scan results were later used for risk assessment and prioritization in the next phase.

Part 4: Risk Assessment and Vulnerability Prioritization

4. Introduction

- After identifying vulnerabilities using OpenVAS in the previous phase, a risk assessment was performed to understand the severity and impact of the discovered issues. Risk assessment helps in prioritizing vulnerabilities based on how likely they are to be exploited and the damage they can cause. This ensures that critical issues are addressed before lower-risk findings.

4.1 Risk Assessment Methodology

- The risk assessment was carried out using the following factors:
 - **CVSS Score:**
Used to measure the technical severity of each vulnerability.
 - **Likelihood:**
Represents how easily a vulnerability can be exploited. Vulnerabilities with default credentials or remote access were considered high likelihood.
 - **Impact:**
Represents the potential damage if the vulnerability is exploited, such as unauthorized access, data leakage, or full system compromise.
- Based on these factors, a final risk level was assigned to each vulnerability.

4.2 Risk Assessment Table

- The following table summarizes the risk evaluation of selected vulnerabilities identified during the OpenVAS scan on the target system (192.168.177.134).



Vulnerability Name	Affected Port/ Service	CVSS Score	Severity(Open VAS)	Likelihood	Impact	Final Risk Level
MySQL Default Credentials	3306/MySQL	9.8	Critical	High	High	Critical
Apache Tomcat Default Credentials	8180/Tomcat	9.8	Critical	High	High	Critical
PHP CGI RCE (CVE-2012-1823)	80/HTTP	10.0	Critical	High	High	Critical
OpenSSH Weak Configuration	22/SSH	7.5	High	Medium	High	High
Outdated Apache Web Server	80/HTTP	7.2	High	Medium	Medium	High

Figure 16: Risk assessment table prepared based on OpenVAS scan results

4.3 Analysis of High-Risk Vulnerabilities

- Several vulnerabilities were classified as **Critical** due to their high CVSS scores and the possibility of remote exploitation.
 - **PHP CGI Remote Code Execution:**
This vulnerability allows attackers to execute system commands remotely, leading to complete system compromise.
 - **Apache Tomcat Default Credentials:**
Default credentials can allow attackers to access the management interface and deploy malicious applications.
 - **MySQL Default Credentials:**
Unauthorized database access can result in data theft or manipulation.
 - **vsftpd Backdoor Vulnerability:**
This vulnerability contains a known backdoor that can provide attackers with shell access.
- These vulnerabilities were given the highest priority for remediation.

4.4 Medium and Low-Risk Findings

- Medium and low-risk vulnerabilities included outdated services, weak configurations, and information disclosure issues. While these vulnerabilities may not lead to immediate system compromise, they can still assist attackers during reconnaissance and should be fixed as part of regular security maintenance.

4.5 Risk Prioritization

- Based on the assessment, vulnerabilities were prioritized as follows:
 1. Remote Code Execution and Default Credential issues
 2. Services allowing unauthorized access



3. Outdated and misconfigured services
 4. Informational and low-impact findings
- This prioritization helps security teams focus on the most dangerous threats first.

4.6 Conclusion of Risk Assessment

The risk assessment phase successfully categorized vulnerabilities based on severity, likelihood, and impact. This step provided a clear understanding of which vulnerabilities pose the greatest risk to the system. The results of this assessment support informed decision-making for remediation and highlight the importance of risk-based vulnerability management.

Conclusion

This VAPT task provided practical exposure to the complete vulnerability assessment and penetration testing process. By setting up a controlled lab environment, performing vulnerability scanning using OpenVAS, and analyzing the results through risk assessment, I gained a clear understanding of how security weaknesses are identified and prioritized in real-world scenarios. The task helped me understand the importance of structured testing, proper tool usage, and systematic documentation in security assessments. Overall, this exercise strengthened my foundation in VAPT and highlighted the role of proactive security testing in reducing organizational risk.

Key Learnings

- Gained hands-on experience in setting up a VAPT testing environment using Kali Linux and Metasploitable 2.
- Learned how to perform vulnerability scanning using OpenVAS and analyze findings based on CVSS scores and severity levels.
- Understood the importance of risk assessment, prioritizing vulnerabilities, and documenting findings in a professional security report.

References

1. OpenVAS / Greenbone Documentation – <https://www.greenbone.net>
2. Nmap Official Documentation – <https://nmap.org/book/>
3. OWASP Top 10 – <https://owasp.org/www-project-top-ten/>
4. CVSS Scoring System – <https://nvd.nist.gov/vuln-metrics/cvss>
5. Metasploitable 2 Project – <https://sourceforge.net/projects/metasploitable/>
6. Kali Linux Documentation – <https://www.kali.org/docs/>